Light

# GCSE Computer Science - Cyber Security Threats (Set 2)

Over **1,100,000+ marks** of exam-aligned content across **67 qualifications**. 4,200+ auto-marked assessments & long-answer practice. No login required.

GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

00:00 Start Pause Reset

← Back to Cyber Security Threats

Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is the primary purpose of adware?

- ○ **A.** To display unwanted advertisements and generate revenue for the creator
- ○ **B.** To encrypt files and demand ransom
- ○ **C.** To spread across networks automatically
- ○ **D.** To record keyboard inputs

Q2 (1 mark)

What makes a rootkit particularly dangerous?

- ○ **A.** It only affects the roots of file directories
- ○ **B.** It hides itself and other malware from detection by security software
- ○ **C.** It can only infect computers with administrator accounts
- ○ **D.** It spreads through plant-related websites

Q3 (1 mark)

What is a botnet?

- ○ **A.** A network of security robots that protect computers
- ○ **B.** A type of antivirus software
- ○ **C.** A network of compromised computers controlled remotely by an attacker
- ○ **D.** A social media platform for sharing bot-related content

Q4 (1 mark)

What is vishing?

- ○ **A.** Phishing conducted over voice calls (telephone)
- ○ **B.** A type of computer virus
- ○ **C.** A method of encrypting voice messages
- ○ **D.** A technique for speeding up internet connections

Q5 (1 mark)

What is smishing?

- ○ **A.** A method for compressing files
- ○ **B.** A type of network topology
- ○ **C.** Phishing conducted via SMS text messages
- ○ **D.** A technique for improving smartphone performance

Q6 (1 mark)

What is baiting in cyber security?

- ○ **A.** Testing network speed with large files
- ○ **B.** Leaving infected physical media (like USB drives) for victims to find and use
- ○ **C.** Sending multiple emails to test spam filters
- ○ **D.** Creating backup copies of important data

Q7 (1 mark)

What is tailgating (or piggybacking) in physical security?

- ○ **A.** Following authorised personnel through secure doors without proper credentials
- ○ **B.** Driving closely behind delivery vehicles

- ○ **C.** Copying someone's password by watching them type
- ○ **D.** Sending follow-up emails after phishing attempts

Q8 (1 mark)

What is shoulder surfing?

- ○ **A.** A water sport using computer equipment
- ○ **B.** Observing someone's screen or keyboard to steal information like passwords or PINs
- ○ **C.** A method of network packet analysis
- ○ **D.** Surfing the internet while standing

Q9 (1 mark)

What is a dictionary attack?

- ○ **A.** Using a list of common words and passwords to try to guess credentials
- ○ **B.** Hacking into online dictionary websites
- ○ **C.** Translating passwords into different languages
- ○ **D.** Looking up the definition of technical terms

Q10 (1 mark)

What is a man-in-the-middle (MITM) attack?

- ○ **A.** An attack where someone physically stands between two computers
- ○ **B.** An attack where the attacker secretly intercepts and potentially alters communication between two parties
- ○ **C.** An attack that only targets middle management employees
- ○ **D.** An attack that splits files in half

Q11 (1 mark)

What is pharming?

- ○ **A.** Redirecting users from legitimate websites to fraudulent ones by manipulating DNS records
- ○ **B.** Growing malware in computer laboratories
- ○ **C.** A method of improving agricultural technology
- ○ **D.** Storing data on multiple servers

Q12 (1 mark)

What is an insider threat?

- ○ **A.** A threat from employees or others with authorised access who misuse their privileges
- ○ **B.** A threat that only affects the inside of computer cases
- ○ **C.** A weather warning for indoor data centres

- ○ **D.** A threat from competitors in the same industry

Q13 (2 marks)

A group of hackers targets a government website to protest against a new policy, defacing it with political messages.
What is this type of attack motivation called?

- ○ **A.** Financial gain - they want money from the government
- ○ **B.** Industrial espionage - stealing government secrets
- ○ **C.** Script kiddie activity - beginners practising hacking
- ○ **D.** Hacktivism - using hacking to promote political or social causes

Q14 (2 marks)

An elderly person receives a phone call from someone claiming to be from their bank's fraud department. The caller says suspicious activity has been detected and asks for account details to 'verify identity'.
What type of attack is this?

- ○ **A.** A brute-force attack on the phone network
- ○ **B.** Smishing - because it involves communication
- ○ **C.** Vishing - phishing conducted via voice call to obtain sensitive information
- ○ **D.** Pharming - redirecting calls to fake numbers

Q15 (2 marks)

A company employee finds a USB drive in the car park labelled 'Salary Information - Confidential'. Curious, they plug it into their work computer.
Why is this action dangerous?

- ○ **A.** USB drives can damage the USB port physically
- ○ **B.** The drive could contain malware that executes automatically when connected, infecting the computer and potentially the network
- ○ **C.** USB drives slow down computer processors
- ○ **D.** The salary information might be incorrect

Q16 (2 marks)

An attacker uses a list of the 10,000 most commonly used passwords to try to access user accounts.
How does this dictionary attack differ from a brute-force attack?

- ○ **A.** Dictionary attacks use targeted word lists and are faster; brute-force tries every possible combination systematically and takes longer
- ○ **B.** Dictionary attacks are legal while brute-force attacks are illegal
- ○ **C.** Brute-force attacks only work online while dictionary attacks only work offline
- ○ **D.** There is no difference between the two attack types

Q17 (2 marks)

A user connects to a coffee shop's Wi-Fi to access their online banking. Unknown to them, an attacker has set up a fake Wi-Fi hotspot with the same name.
What type of attack could the attacker now perform?

- ○ **A.** SQL injection - inserting code into the coffee shop's database
- ○ **B.** Ransomware - encrypting the user's phone
- ○ **C.** Man-in-the-middle - intercepting all traffic between the user and banking website
- ○ **D.** Brute-force - guessing the user's Wi-Fi password

Q18 (2 marks)

A disgruntled employee who has just been made redundant copies sensitive customer data to a personal USB drive before leaving the company.
What type of threat does this represent?

- ○ **A.** External hacking - the employee is now outside the company
- ○ **B.** Social engineering - manipulating others to gain access
- ○ **C.** Malware infection - the USB drive contains viruses
- ○ **D.** Insider threat - an authorised user misusing their legitimate access to steal data

Q19 (2 marks)

A computer has been running slowly for months. Antivirus scans show no infections, but the user suspects malware.
Why might a rootkit be responsible and undetected?

- ○ **A.** Rootkits only slow down computers, they do not show in scans
- ○ **B.** Antivirus software cannot scan computers that run slowly
- ○ **C.** Rootkits hide themselves by modifying the operating system, making them invisible to standard antivirus software

○ **D.** Slow computers automatically disable antivirus scanning

Q20 (2 marks)

A user types their bank's correct web address into the browser but is redirected to a fake website without clicking any links.
What type of attack is this, and how does it differ from phishing?

○ **A.** This is standard phishing - all phishing redirects users automatically

○ **B.** This is pharming - it manipulates DNS to redirect users without requiring them to click malicious links, unlike phishing which relies on deceptive links

○ **C.** This is a brute-force attack on the bank's website

○ **D.** This is impossible - typing correct addresses always reaches correct sites

Q21 (2 marks)

A cybercriminal controls 50,000 compromised computers around the world. They instruct all these computers to simultaneously send requests to a single website.
What attack are they launching and why are botnets effective for this purpose?

○ **A.** A virus attack - the computers are spreading infections

○ **B.** A phishing attack - sending emails from many computers

○ **C.** A SQL injection attack - using multiple computers to access databases

○ **D.** A DDoS attack - the distributed botnet provides massive traffic volume from many sources, overwhelming the target while making

blocking difficult

Q22 (2 marks)

An attacker wearing a delivery uniform and carrying boxes waits by a secure office entrance. When an employee badges in, the attacker says 'Could you hold the door? My hands are full.'
What attack technique is being used?

- ○ **A.** Brute-force - trying to break through the door
- ○ **B.** Phishing - sending a deceptive message
- ○ **C.** Tailgating - following an authorised person through a secure entrance by exploiting social norms
- ○ **D.** SQL injection - manipulating the door access system

Q23 (2 marks)

Attackers discover a previously unknown security flaw in popular software and begin exploiting it before the software company is aware of the problem.
What is this type of vulnerability called?

- ○ **A.** A legacy vulnerability - it has existed for a long time
- ○ **B.** A zero-day vulnerability - it is unknown to the vendor and has no patch available
- ○ **C.** A patched vulnerability - it has already been fixed
- ○ **D.** A theoretical vulnerability - it cannot actually be exploited

Q24 (3 marks)

An attacker uses the following sequence: They call a company receptionist pretending to be IT support, obtain an employee's email address, send that employee a targeted phishing email with malware, then use the infected computer to access the company database. Analyse the attack techniques used at each stage.

- ○ **A.** All stages used the same technique - standard hacking
- ○ **B.** The entire attack was automated by artificial intelligence
- ○ **C.** This describes normal IT department operations
- ○ **D.** Stage 1: vishing (voice social engineering), Stage 2: spear phishing (targeted email attack), Stage 3: malware infection, Stage 4: data breach using the compromised system as entry point

Q25 (3 marks)

A security trainer needs to explain the differences between phishing, vishing, smishing, and spear phishing to employees.
Analyse the key distinguishing characteristics of each attack type.

- ○ **A.** They are all identical attacks with different names
- ○ **B.** Phishing uses email; vishing uses voice calls; smishing uses SMS texts; spear phishing is any of these but highly targeted at specific individuals using personalised information
- ○ **C.** These attacks only affect different types of devices - computers, phones, and tablets respectively
- ○ **D.** Only phishing is a real threat; the others are theoretical

Q26 (3 marks)

A company experiences three incidents: an employee accidentally emails confidential data to the wrong recipient, a system administrator

intentionally sells customer data, and a contractor's login credentials are stolen by external hackers.
Analyse how each incident relates to insider threats.

- ○ **A.** None of these are insider threats - they all involve different people
- ○ **B.** Only the intentional sale is an insider threat; accidents do not count
- ○ **C.** All three are examples of malware infections
- ○ **D.** The accidental email is a negligent insider threat, the intentional sale is a malicious insider threat, and the stolen credentials enable external attackers to pose as insiders - all represent different insider threat categories

Q27 (3 marks)

Three separate cyber attacks occur: a ransomware attack demanding Bitcoin payment, an attack stealing trade secrets from a competitor, and website defacement protesting environmental policy.
Analyse the likely motivation behind each attack.

- ○ **A.** All three attacks have the same motivation - causing disruption
- ○ **B.** Ransomware: financial gain (profit from payment); trade secret theft: industrial espionage (competitive advantage); website defacement: hacktivism (political/social protest)
- ○ **C.** All attackers are motivated by boredom and testing their skills
- ○ **D.** These attacks have no motivations - they happen randomly

Q28 (3 marks)

A security consultant is explaining how HTTPS protects against man-in-the-middle attacks when using public Wi-Fi.

Analyse how the different components of HTTPS provide this protection.

- ○ **A.** HTTPS only makes websites load faster, it does not provide security
- ○ **B.** HTTPS blocks all connections from public Wi-Fi networks
- ○ **C.** HTTPS only protects against viruses, not man-in-the-middle attacks
- ○ **D.** HTTPS uses encryption to make intercepted data unreadable, and certificate verification confirms website identity - together preventing attackers from reading data or impersonating legitimate sites

Q29 (3 marks)

Attackers want to spread malware to as many computers as possible. They consider using viruses, worms, or Trojans.
Analyse the advantages and limitations of each distribution method for achieving maximum spread.

- ○ **A.** All three methods spread malware in exactly the same way
- ○ **B.** Only Trojans can spread malware; viruses and worms are not used for distribution
- ○ **C.** Modern systems are immune to all three distribution methods
- ○ **D.** Viruses spread via infected files (relies on sharing); worms self-propagate across networks (fastest autonomous spread but requires vulnerabilities); Trojans need social engineering (highest success rate per target but slowest scale)

Q30 (3 marks)

A small business has the following setup: employees use personal devices for work, the company uses cloud services accessed via web

browsers, staff connect from home and coffee shops, and the website accepts customer form submissions.
Analyse the potential attack vectors this creates.

- ○ **A.** Small businesses cannot be targets of cyber attacks
- ○ **B.** This setup has no security risks because cloud services are inherently secure
- ○ **C.** Only large enterprises need to consider attack vectors
- ○ **D.** Personal devices create BYOD risks (uncontrolled security); cloud access via browsers risks credential theft; remote connections from untrusted networks enable interception; web forms create SQL injection and input-based attack surfaces

Total: **55** marks • Time: **30 min**

↑ Back to top

Submit answers

**Williams Physics Education** — over **1,100,000+ marks** of exam-aligned content and **4,200+ auto-marked assessments** across 67 qualifications at GCSE, A-Level, AP & Professional Qualifications. Free to use, no login required.

About & Contact • Privacy • Terms • Accessibility • Copyright • DMCA • YouTube

WPE is free for everyone. If our resources help you or your students, you can support the project here.

© 2026 Williams Physics Education. All rights reserved.